MILITARY SPECIFICATION

ELECTRODES, WELDING, MINERAL COVERED, LOW-HYDROGEN, MEDIUM AND HIGH TENSILE STEEL

This amendment forms a part of Military Specification MIL-E-22200/6C, dated 15 March 1976, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, under "STANDARDS" add:

"FEDERAL
FED-STD-151 - Metals, Test Methods."

PAGE 2

2.2: Add:

"DEPARTMENT OF LABOR
Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)"

PAGE 3

Add as paragraphs 3.6, 3.6.1, and 3.7:

"3.6 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-22200.

"3.6.1 Heat and lot identification. When specified (see 6.2), each electrode shall be marked with heat or lot identification as specified in 3.5.10 of MIL-E-22200.

"3.7 Alloy identity. The alloy identity provisions of 3.6.4 of MIL-E-22200 shall be applied to each length of rod or wire to be spliced and to both ends of each wire coil drawn to finished size prior to cutting into electrodes. When specified (see 6.2), the core wire of each covered electrode shall meet the alloy identity requirements of 3.6.5 of MIL-E-22200."

PAGE 5

Table X: Add:

"Tests	Procedures	Requirements
Alloy identity	4.6.11 (MIL-E-22200)	3.6.4 (MIL-E-22200)"

FSC 3439

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PAGES 5 and 6

4.6: Delete and substitute:

"4.6 Certification of quality conformance. A certificate of quality conformance of the weld tests specified in 4.4.3 (i.e., X-ray, mechanical, and chemical analysis), containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot, heat, and wet mix, as applicable, included in a particular shipment shall be furnished to the consignee with the shipment. The certification shall include test results from the quality conformance test, alloy identity test, and the wet mix equivalency test results as applicable for the shipment. The minimum quality conformance test data, alloy identity test data and wet mix equivalency test result data required shall be as shown on figures 3 and 4 and may be submitted in the format of figures 3 and 4."

PAGE 6

Add as paragraphs 4.6.1 and 4.6.2:

"4.6.1 The identification of melts, heats or lots and the general requirements for test specimens shall be in accordance with FED-STD-151.

"4.6.2 Alloy identity certification. When specified (see 6.2), the inspector shall affix his stamp and date on the container, containing electrodes conforming to 3.6.5 of MIL-E-22200. The letters 'AT' shall be affixed to the container adjacent to the date to indicate 'Alloy Tested'."

5.3.1.1.1, CAUTION label: Delete and substitute:

"

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

Read and understand the manufacturer's instructions and your employer's safety practices.
Keep your head out of the fumes.
Use enough general ventilation or exhaust at the arc or both to keep fumes and gases from your breathing zone, and the general area.
Wear correct eye, ear and body protection.
Do not touch live electrical parts.
See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, 550 N.W. LeJeune Road P.O. Box 351040, Miami, Florida 33135; OSHA Safety and Health Standards, 29 CFR 1910 are for sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

DO NOT REMOVE THIS LABEL

"

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PAGE 7

Add as paragraph 5.3.1.3:

"5.3.1.3 Containers shall be marked 'ASBESTOS FREE'."

6.1: Delete and substitute:

"6.1 Intended use. Degradation of high strength weld metal tensile and impact properties may occur because of exposure to stress relief temperatures and cooling from stress relief. These mechanical properties should be redetermined after simulated production heat treatment of sample material if required by applicable welding specifications and standards."

6.1.1: Add "and composition C of MIL-S-23194 and MIL-S-24238".

6.1.2: Add "composition H and A of MIL-S-23194 and composition A of MIL-S-24238".

6.2: Add as items (k) and (l):

"(k) Whether heat or lot identification is required for individual electrodes (see 3.6.1).
(l) Whether the alloy identity test and certification is required for the core wire of each finished electrode (see 3.7 and 4.6.2)."

PAGE 9

Figure 1, notes 6 and 8: Delete and substitute:

"6. One tensile coupon, where required by the acquisition document, shall be stress-relieved using the following procedure:

(a) Employing an electric furnace, the all-weld metal coupon shall be heated to 1125°F ± 25°F (for 7015 and 7016), 1150°F plus 25°F minus 50°F (for 8015 and 8016), or 1075°F ± 25°F (for 10015 and 10016), held for 2 hours at temperature and allowed to furnace cool at a rate not to exceed 100°F per hour maximum to a temperature below 500°F."

"8. The center section of the welded joint shall be used for the preparation of charpy V-notch impact specimens. The charpy V-notch specimens shall be machined in accordance with details shown on figure 1. Prior to machining specimens, where required by the acquisition document, the coupon shall be subjected to the following stress-relieving condition:

(a) Using an electric furnace, the weld coupon shall be held for 2 hours at 1125°F ± 25°F (for 7015 and 7016), 1150°F plus 25°F minus 50°F (for 8015 and 8016), or 1075°F ± 25°F (for 10015 and 10016), and then allowed to cool to a temperature below 500°F at a rate not to exceed 100°F per hour."

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Figure 3: Delete and substitute the attached Figure 3.

LAST PAGE

DD 1426, Standardization Document Improvement Proposal: Delete and substitute address:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362"

Custodians:
Army - AR
Navy - SH
Air Force - 99
Review activities:
Army - MR
Navy - OS
DLA - GS
User activities:
Army - AT, ME
Navy - MC, YD

Preparing activity:
Navy - SH
(Project 3439-0405)

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CERTIFICATION OF QUALITY CONFORMANCE

Manufacturer or distributor _____ Customer's name _____

Address _____

Date _____ Customer's order no. _____

Specification MIL- _____

Type MIL- _____ Core wire heat no. _____

Diameter and length _____ Lot identification MIL-E-22200 Para. _____

Inspection level _____

Lot no. _____ Wet mix no. _____

Chemical analysis (Complete)	Yield strength (0.2 percent offset method)	Mechanical tests As-welded	Mechanical tests Stress-relieved
Carbon _____	Tensile strength	_____	_____
Manganese _____	Elongation (percent in __ inches)	_____	_____
Silicon _____		_____	_____
Phosphorus _____	Reduction of area (percent)		
Sulfur _____		Charpy impacts	
Chromium _____		1. _____	1. _____
Nickel _____		2. _____	2. _____
Molybdenum _____		3. _____	3. _____
Vanadium _____		4. _____	4. _____
Copper (RC electrode) _____		5. _____	5. _____

Chemistry was taken from: ☐ Chem pad ☐ Groove weld

X-Ray results _____ Groove weld test

Concentricity (percent) _____

Test no.	3	8a	Chem pad
Amperage	_____	_____	_____

Covering moisture _____

FIGURE 3. Certification of quality conformance tests.

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Grinding During 8a Test Plate Preparation (see note 3)
Operator Error (Layer Nos.) _____

Alloy identity tests:

(a) Core wire _____

(b) Finished electrodes _____(if required, see 3.7)

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Authorized contractor representative

NOTES:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. Figure 4 forms shall be employed to report and certify the wet mix equivalency tests where applicable for each wet mix of covering mixture in the lot. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. See paragraphs 3.5.7 and 3.9 of MIL-E-22200.
4. See 4.6 herein.

FIGURE 3. Certification of quality conformance tests. - Continued

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